Categorizing Institutional Logics, Institutionalizing Categories:

A Review of Two Literatures

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Abstract

This review assembles two highly referenced streams of research in organization and management studies over the past decade: institutional logics and categories. We present the gist of each literature focusing on the interaction within and between organizations vis-à-vis the institutional logics and category systems that condition behavior. Then, we suggest that both streams have compatible assumptions that warrant further integration and suggest opportunities for future research stemming from (1) complementarities related to inter- and intra-audience variance, formation and recombination of logics and categories, and actors' identity and (2) differences related to theory level of analysis, incorporation of conflict, and methods of analysis. Integration can lead to better specified mechanisms, processes, and contexts important to improving accuracy and development of these research streams.

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The field of organization and management studies has developed into fragmented parallel streams of research. For scholars interested in institutional change, organizational behavior, and decision-making within organizations, it is time to assess the opportunities for converging existing frameworks to enhance scientificity, accumulate findings, and improve theorizing (Okhuysen and Bonardi, 2011; Cornelissen and Durand, 2012). This review analyzes the reasons and challenges for rapprochement between the institutional logics and categories research, which share positive momentum in recent publication numbers, a common research community, and focus on similar questions and problems.

A record of publications, tripling in a decade the number of yearly articles published in top journals, identifies these two perspectives as vibrant research streams within organization and management studies. Since 2011, approximately 120 articles using the institutional logics framework appeared in top journals each year in the fields of communications, management, strategy, entrepreneurship, education, public policy, and sociology (Ocasio, Thornton, and Lounsbury, 2017). About 100 articles on average for market categories appeared in top journals in psychology, marketing, strategy, organization theory, economics, and sociology (Durand and Paolella, 2013; Vergne and Wry, 2014; Durand, Granqvist, and Tyllstrom, 2017). Many of the same authors contribute to both domains, however in an uncoordinated fashion thus far. For instance, Greenwood examined how the influences of multiple institutional logics moderate organizational decision making and more recently recategorization and gains in status (Greenwood et al, 2010; Delmestri & Greenwood, 2016); Lounsbury studied the effects of competing institutional logics on practice adoption and recently nascent categories (Lounsbury, 2007; Wry & Lounsbury, 2011; Lee, Hiatt & Lounsbury, 2017); Durand argued in favor of a refocus of research on goal-based categories and examined how institutional logics can be used as strategic resources (Durand and Paolella, 2013; Durand, Szostak, Jourdan & Thornton, 2013). Finally, both perspectives tackle fundamentally the same questions: what is the value of deviance vs. conformity? Why and how do individual and organizational actors maintain or upend preexisting orders? Why do certain organizations succeed whereas others decay and disappear?

The institutional logics perspective is defined as a framework for analyzing the interrelationships among individuals, organizations, and institutions in social systems (Friedland and Alford, 1991; Thornton, Ocasio, and Lounsbury, 2012). Institutional logics are "the socially constructed historical patterns of cultural symbols and material practices, assumptions, values, and beliefs by which individuals produce and reproduce their material subsistence, organize time and space, and provide meaning to their daily activity" (Thornton and Ocasio, 1999: 804). Researchers have shown that institutional logics have influences on actors at different levels of analysis, i.e., individuals, groups, and organizations, for example with respect to executive succession (Thornton and Ocasio, 1999), corporate governance (Geng, Yoshikawa and Colpan (2016); corporate networks (Bhappu, 2000), practice adoption (Lounsbury, 2007), professional identity (Dunn and Jones, 2010), organizational deviance (Durand and Jourdan, 2012), organizational foundings (Almandoz, 2012), corporate social responsibility (Zhang and Luo, 2013; Ioannou and Serafeim, 2015), firm innovation (Pahnke, Katilla and Eisenhardt, 2015) and interorganizational learning (Vasudeva, Alexander, and Jones, 2014), among others.

Categories are defined as interfaces of cognitive agreement about a considered object. More precisely, categories are the symbolic and material attributes of products, firms, and industries that are both shared among actors and that distinguish these entities from others (Carruthers & Stinchcombe, 1999; Douglas, 1986; Zerubavel, 1991; Zuckerman, 1999). Market actors, both producers and consumers, use categories to represent themselves and to communicate with each other in markets for products and services. A high level of agreement about what it means to be a member of the set (e.g., a craft brewer, a comedy movie) suggests that the category is institutionalized and taken for granted (Hannan Polos, and Carroll, 2007; Negro, Kocak, and Hsu, 2010). Without categories serving as the "social agreements about the meanings of labels applied to them" (Negro, Hannan and Rao, 2011: 1450), neither institutional consensus nor social, economic, and symbolic exchanges are likely to be achieved. Research provides evidence of the influence of categories on competitors' recognition (Porac et al, 1989; 1999), organizational evaluation and reputation (Hsu et al, 2009), strategic opportunities (Hsu and Grodal, 2015), and firm performance (Paolella and Durand, 2016).

A few researchers have previously connected the two literatures, but there has not been a systematic review for this purpose. For instance, Ocasio, Lowenstein, and Nigam, (2015) suggest that categories constitute institutional logics through the distribution of streams of communication in organizations and institutional fields. As evidence continues to accumulate that these two perspectives are fertile terrains for organization and management research (Lounsbury and Beckman, 2015; Durand et al, 2017), and multiple scholars engage in both literatures, it makes sense to further highlight promising areas of integration and cross-fertilization. For example, the institutional logics stream fails to mobilize useful concepts from the category research, such as prototypes, goal-based categories, and theories of value. Similarly, the categories stream sometimes refers to an institutional logic but remains evasive as to how precisely institutional logics influence outcomes.

What do we miss if we do not integrate these two perspectives? What is to be gained? How do mutual oversights create needless shortcomings in both literatures? To address these questions, the purpose of this review is to identify and present the core ideas of each research stream (section I), and to examine the epistemological compatibility, and complementarities and differences across both streams in order to suggest avenues for future research (section II). We detail six avenues related to inter- and intra-audience variance, formation and recombination of logics and categories, actors' identity, theory integration and boundary conditions, incorporation of conflict, and use of multiple methods.

I. LITERATURE REVIEW: KEY IDEAS AND CONCEPTS

First, we review articles published in the top journals[1] as well as significant contributions from other impactful journals, edited volumes, and books. Second, we rely on knowledge of the two literatures over the last two decades that is published separately elsewhere (Thornton and Ocasio, 2008; Thornton, Lounsbury, Ocasio, 2012; Ocasio, Thornton and Lounsbury, 2017; Durand and Paolella, 2013; Durand et al. 2017). Third, this review is articulated in two subsections, beginning with how institutional logics influence organizations, how organizations deal with multiple institutional logics, and how individual and organizational actors instigate institutional change. Then, the review details the research on the formation of categories, the antecedents and consequences of category membership, and how actors strategically develop category memberships.

I.1. From Organizational Fields to Institutional Logics

Friedland and Alford's (1991) critique of institutional and organization theory provided the foundation for a new perspective in institutional theory. In essence, Friedland and Alford (1991) melded Marxian conflict theory with Weberian value spheres to identify solutions to long

[1] Administrative Science Quarterly; American Journal of Sociology; American Sociological Review; Academy of Management Journal; Academy of Management Review; Journal of Management Studies; Management Science; Organization Science; Strategic Management Journal

standing problems of agency, bounded rationality, self-interest, power, and resources in organization and management studies. While this approach was not new (Weber, 1978; Merton, Brown, and Cottell, 1959), it had not been applied to contemporary institutional and organization theory. More precisely, Friedland and Alford (1991) countered that society is an inter-institutional system, where for example the family, religion, market and the state coexist as institutional orders, and that each institutional order differentially influences individuals' and organizations' rationality and actions. Moreover, the differences in organizing principles across institutional orders heighten actors' attention to contradictions which create opportunities to manipulate institutional logics for purposeful gain with implications for institutional change. Fundamentally, each institutional order is a cultural knowledge corridor that distinctly shapes attention and the interpretation of meaning.

Building on this critique, scholars empirically demonstrated the limits of neo-institutional theory in a number of ways; its inability to explain adaptation (Kratz and Zajac, 1996), its lack of a theory of action (DiMaggio, 1988; Hardy and Mcguire, 2008), its over emphasis on the structural influence of an organizational field (Thornton and Ocasio, 1999), its singular focus on mimetic isomorphism (Mizruchi and Fein, 1999), and its dependence on two-stage models of practice diffusion (Lounsbury, 2007; Purdy and Gray, 2009; Shipilov, Greve and Rowley, 2010). Simplistic views of organizational actors as rational (motivated by efficiency) or non-rational (motivated by legitimacy) based on market (technical) versus normative (institutional) environments (Meyer and Rowan, 1977), were ripe for revision. We highlight in subsequent subsections the themes characterizing this revision.

Influence of Institutional Logics on Organization Decisions. An early series of studies showed that a change in the dominant institutional logic changed the determinants of organizational decision making (Thornton, 2004). Developed in the higher education publishing industry, these studies integrated the institutional logics perspective with the Carnegie School of attention (Ocasio, 1997) to examine the determinants of CEO selection (Thornton and Ocasio, 1999), risk of acquisition (Thornton, 2001), and adoption of organizational form (Thornton, 2002). Key findings were a change from family and professional to market institutional logics changed the basis of attention to highlight different dimensions of corporate governance.

How institutional logics affect organizational decisions and outcomes continues to be explored in conjunction with a variety of organization theories, evidencing different mechanisms for organizational creation and strategic choice. For instance, economic cycles and the level of association of founders' functional backgrounds affect how opportunity structures appear to organizations and their leaders and predict the founding of banks (Almandoz, 2012). Bank management teams with a greater proportion of members with a finance background (market logic) as distinct from a community background (community logic), were more likely to use risky deposit instruments to spur rapid growth (Almandoz, 2014). Marquis and Lounsbury (2007) linked ecological theory to institutional logics to show how conflicting logics seeded in local (community) versus global (market) principles inspired a resistance movement. They demonstrated how the acquisition of local banks (specialists) by out of town large banks (generalists) fueled an increase in the founding of small community banks. That is, national banks' efforts to introduce global efficiencies by geographic diversification ironically triggered resource partitioning and entrepreneurialism to preserve the community logic in banking. Integrating ecology but also category theory Jourdan (forthcoming) finds that firms that are specialized to an institutional logic have organizational survival advantages which increase when the evolution of conflicting institutional logics in the industry (population) decrease. Greve and

Zhang (2017) linked institutional logics to firm power coalitions to examine firm strategic choice. In the context of Chinese firms, the strength of the effects of firm coalitions changed as the market logic gained dominance. Not surprisingly, investors were skeptical of firms with high state ownership engaging in the strategy of mergers and acquisitions--as it is consistent with a market, not a state institutional logic.

Organizational decisions however may not depend on differences in one institutional logic versus another, but instead on the effects of a constellation of institutional logics (Goodrick and Reay, 2011), such as in the cases of innovation in venture-backed firms (Pahnke, Katilla, and Eisenhardt, 2015) and innovation in product design (Dalpiaz, Rindova, and Ravasi, 2016). Greenwood, Diaz, Li and Lorente (2010) for example investigated the influences of multiple institutional logics on corporate downsizing in Spanish manufacturing firms, a strategy motivated by the market logic. In regions where the influence of the Catholic church (religion logic) was strong, with its emphasis on family values, the strategy of corporate downsizing was moderated, even during economic down turns. This study is important because moderation in the adoption of orthodox market principles did not result solely from the typical source of state regulation, but from the influences of other institutional orders, in their study, religion. Building on the idea of one logic constraining or moderating the effects of another, Lee and Lounsbury (2015) examined how the community logic, operationalized as differences in conservative and liberal political ideology, amplified or dampened the influences of state and market institutional logics in shaping firms' toxic waste emissions. In a rare international comparison case, Kuipers (2015) demonstrated how screen translators aired the same television show, but strategically varied its contents for audiences in France, Italy, Netherlands and Poland. The key point is that in spite of the globalization of the media industry, national norms persisted because translators' guiding institutional logics were not determined solely by transnational global media companies, but moderated by technological and organizational differences that varied by national contexts.

Conflicting Institutional Logics Within Organizations. In parallel to the first stream of research demonstrating the *consequences* of multiple logics *on* organizational decisions, another vein of research examines organization and management processes to solve the problems created by competing and conflicting institutional logics *within* organizations (Besharov and Smith, 2014). Notably, Battilana and Dorado (2010) examined micro-finance, an organizational form to exploit new market opportunities to sell banking services to the poor. The bank recruiting employees with no prior experience in either the market or the state welfare logic was more successful than the bank hiring employees with prior experience in either finance or social work. This clean slate approach was more successful because managers socialized individual employees early on from the ground up to value integration of both logics and therefore mitigated the conflict that interfered with productive working relationships. Micro-finance also has been investigated in terms of how the variable gender interacts with different institutional orders to support or suppress outreach to marginal categories of people (Zhao, Yanfei, and Wry, 2016). Moreover, banks' ability to raise funds for micro-finance lending differs by the type of institutional logic associated with the fund, e.g., financial (market) and development (state) logics (Cobb, Wry and Zhao, 2016).

Also, within the social welfare domain, Pache and Santos's (2013) showed how non-profit organizations managed conflicting logics by selectively coupling intact elements of different logics. Over identification with the commercial logic, a strategic misfit, diminished legitimacy and motivated organizations to adopt "Trojan horse" practices by incorporating categorical elements from the social welfare logic to offset legitimacy losses. In another case of logic misfit related to the public-private Cambridge Energy alliance, Jay, (2013) showed that logic contradiction creates paradoxes that motivate actors to create hybrid organizational forms. Paradox is defined in the sense that the organization was deemed a success when viewed through the lens of one logic and a failure when viewed through the lens of another logic.

Hybridization and selective coupling, however are not the only mechanisms to manage conflicting institutional logics. A series of studies examine other mechanisms, for example truces in health care (Reay and Hinings, 2009), geographic distance in financial services (Lounsbury, 2007), and balancing logics in the reinsurance industry (Smets et al. 2015). In Lounsbury's (2007) study of the mutual fund industry, the Boston trustee (professional logic) and the New York performance (market logic) resulted in alternative practices and products whose differences were maintained by geographic separation. In Smets, Jarzabkowski, Burke and Spee (2015) study of reinsurance trading at Lloyd's of London, actors responded to conflicting logics by segmenting, bridging, and demarcating. Actors used these mechanisms to achieve a dynamic balance that rendered institutional logics complementary by maintaining the distinction between conflicting logics, but also retaining the benefits of their interdependence. Individuals' management of co-existing logics became ritualized in a "continual balance of dynamic tension," which dampens incentives for blending or segregating of institutional logics.

Yet, in some cases of conflicting institutional logics, actors must authentically maintain their identity to effectively co-produce. Jourdan, Durand, and Thornton (2017) examined how actors engaged in symbolic interaction and deferent behavior to respond to the strain of conflicting institutional logics. The authors integrate Goffman's (1956) classic theory with institutional logics in a mixed methods study of the population of French film producers and financiers. Financiers traditionally mocked for their focus on money rather than aesthetics are vitally needed for financing films. These financiers cannot employ previously reviewed mechanisms (i.e. hybridize, selectively decouple or switch their identity) as they are stigmatized actors and must perform their functional role of bankers. By expressing deference certain financiers are able to cooperatively position themselves in enemy territory to co-produce films. The investment funds more likely to engage in deferent behavior had more bankers on their board, hence a greater need for "presentation of self" and to symbolically manage their image through deference. Deferent financiers were less inclined to lock-in their identity by making public pronouncements committing to strict financial goals and were caught up in the ritual whirl of the Cannes film festival awards circuit. While, deferent financiers were able to build social capital with film producers, it was only at the price of decrementing profits. This study measured the conditions of capital conversion (from financial to social capital) in a context of conflicting institutional logics.

In sum, this vein of research identifies a variety of empirical instantiations and theoretical mechanisms that organizations employ when hosting multiple and conflicting logics. Hybridization (Batillana and Dorado, 2010), selective coupling (Pache and Santos, 2013), geographic separation (Lounsbury, 2007), balancing (Smets et al, 2015) and deferent behavior and symbolic interaction (Jourdan et al, 2017) form the repertoires of organizational actions.

Influencing Institutional Change. Through their actions, organizations trigger changes in institutional logics and shifts in institutional logics precipitate changes in organizations, opening interesting areas for investigation. Thornton, Jones and Kury (2005) compared three industries, publishing, architecture, and accounting, and revealed differences in the patterns of institutional change. The practices of publishers were shaped by market rationalization based in an evolutionary model; architects were influenced by a duality of professional logics associated with a cyclical model, and accountants exhibited a punctuated equilibrium model activated by economic scandals and cycles and state regulation. This comparative analysis revealed that different types of organizations and industries face different constraints and opportunities in their efforts to undergo and influence institutional change.

Suddaby and Greenwood (2005) documented in an ethnography at the micro level how institutional entrepreneurs used rhetorical strategies for their own benefit to initiate a new organizational form integrating accounting and legal services: They exposed contradictions in logics and framed those contradictions using vocabulary with narrative fidelity with broader templates in the institutional environment to make the innovation palatable to audiences. In their inductive case study of the wind energy field in Colorado, York, Hargrave, and Pacheco (2016) found that social movement organizations (SMOs), electric utility firms, hybrid organizations, and policy makers responded differently to the incompatibility between economic and ecological pressures. When SMOs initially failed to convince utility firms by using economic arguments (market logic), SMOs switched tactics to engage utility firms using ecological arguments (community logic). Once succeeding in convincing utility firms and hence altering the balance of power in the field, hybrid organizations then emerged. In some domains, institutional logics share jurisdiction, albeit in ongoing tension. In Dunn and Jones (2010), the competing institutional logics of care and science continued to coexist as they were identified with two different types of professionals engaged in an ongoing jurisdictional tension to control the medical educational space. The two logics co-existed in varying prevalence to one another over time because of exogenous forces including an increase in the female composition of medical schools, the advent of rival public health programs, and the heightened public discourse on managed care.

What matters in these different views of institutional change, is that actors at any level have some capability to adjust their reference to the dominant institutional logic, therefore affecting its valence in the market or field and opening an alternative for conformity to and deviance from it. For instance, Durand and Jourdan (2012) in an earlier study of investment funds in the French Film industry examined to what extent organizations respond favorably to demands from minority resource suppliers that hold an unconventional institutional logic. They found that a favorable response to actors espousing a minority logic decreases the influence of the dominant actors in the market, alters the social structure of resource suppliers, and promotes the emergence of new institutional logics. In effect, this form of "alternative conformity" is used by firms occupying a minority position as a soft control strategy towards dominate firms in the market. Jones and Livne-Tarandach (2008) through textual analysis identify words that are available in the architect's cultural tool kit that capture the institutional logics of different institutional orders. They found that architectural firms deploy multivalent keywords that bridge institutional logics to allow for multiple meanings so as to be appealing to the interests of multiple and diverse audiences. Lepoutre and Valente (2012) in a multi-case inductive study of the Belgian horticulture industry explain how organizations become desensitized and immune to the symbolic and material carriers of institutional logics, thereby losing their moorings and making organizations prone to engage in a deviant logic.

Pushing further the idea of institutional logics as strategic resources, Durand, Szostak, Jourdan and Thornton (2013) analyze how three institutional logics, formalism, modernism, and managerialism in the French design industry are resources organizations use to leverage strategic choices. Firms with a lower degree of competence and a broader industrial scope are more likely to add the managerialism logic to their strategic repertoire or tool kit. These firms abandon their association with their original institutional logic, becoming purists in managerialized industrial design, which is distinct if not foreign from their artistic and technical roots. Interestingly, high status firms trigger institutional change in the industry by adding the managerialism logic to their repertoire, yet they maintain reference to their original logic (either functionalist or formalist) and by doing so do not undermine their social position and advantage.

As the institutional logics' research agenda evolved, more emphasis has been placed on how institutional heterogeneity allows individual and organizational actors to manipulate the conflicts and complementarities of the categorical elements of the institutional orders of society. Yet, two fundamental precepts of this perspective have been challenged.

The first critique is based on arguments that people have limited capacity to apprehend contradiction, challenging a central mechanism of action in the institutional logics research, that conflict in logics reveals contradictions and paradoxes that motivate action. Voronov and Yorks (2015) identified "blockages and facilitators" of people's capacity to apprehend institutional contradiction. This critique however ignores prior research on attention and sense making that exposes institutional contradictions, for example Nigam and Ocasio (2010) examined event sequencing around the Clinton Health Care reform initiative and how it triggered actors' sense-making. Along the same line, Clemente, Durand, and Roulet (2017) reposition institutional change in the context of the macro-historical context by associating changes in attention, cognition, status, and identity to decisions to combine or switch logics, which lies the groundwork for organizational and individual decisions, labelled ex post as historical events.

The second issue concerns the idea that rationality does not stem from external ordering principles such as institutional orders. Studying the development of spiritual self-accountability, administrative accounting, and record keeping in the Jesuit Order, Quattrone (2015) argues that

Jesuit principles of rationality were unfolding based on "procedural rationality." He defines procedural rationality as continuous interrogations stemming from rhetorical practices used to classify, recall, and invent knowledge. In essence, Quattrone (2015) questions the need for a substantive logic to ascertain the relationships between means and ends and suggests that actors develop the conditions to sanction their later behavior. The generalizability of Quattrone's arguments deserve caution due to the narrow scope conditions of the Catholic Church and the significant gap in historical time between the 16th century and the advent of modern and well-established institutions in society. Yet, the Jesuit case cautions institutional logics researchers to think more seriously about the bottom-up processes of institutionalization and post hoc sense making.

At the same time, investigators are increasingly focusing on bottom up processes and attentional cues prior to decision-making that explain if individuals and organizations follow the principles of institutional logics blindly or not. Experimental designs are emerging in the literature to explore the psychological schema by which societal institutional logics shape action. Glaser, Fast, Harmon and Green (2016) integrate theory from dynamic constructivism in social psychology with the institutional logics perspective, and show that subjects, exposed to a cue that is associated with a particular institutional logic, activate logic consistent schema, making the logic cognitively salient and accessible. For example, subjects primed to cues of a market logic justified past actions as more self-interested relative to subjects exposed to a family logic cue. The authors further demonstrate that subjects exposed to market logic cues exerted more effort on a self-interest task than subjects exposed to a family logic cue.

As the more recent research indicates, the focus of the institutional logics research is moving inside and between organizations in a market to examine how normative impositions combine with cognitive cues and mechanisms to explain behavior. Moreover, it is noteworthy that interest in the cognitive determinants of choices in the context of markets ribbed with institutional logics delineates a clear interface between the institutional logics perspective and the research on categories as subsequently discussed.

I.2. Categories: from types to goals

Categories in the context of markets and organizations "provide a cognitive infrastructure that enables evaluations of organizations and their products, drives expectations, and leads to material and symbolic exchanges" (Durand & Paolella, 2013: 1102). Categories are the symbolic and material attributes of products, firms, and industries that are both shared among actors and that distinguish these entities from others (Carruthers & Stinchcombe, 1999; Douglas, 1986; Zerubavel, 1991). An early harbinger of the categories research is the work on genres and styles in cultural fields that facilitate recognition and assessment (e.g., the study of popular music by Peterson and Kern, 1996). Zuckerman's (1999) research on securities analysts hallmarks the initial theoretical formulation of the categories research, introducing the concepts of the "categorical imperative" and the illegitimacy discount in which firms that fit within identifiable categories are privileged and those that do not are penalized. In the following review, we examine how organization scholars explain category formation, the consequences of belonging to one or many categories, and the strategic behavior of actors in relation to category systems.

Formation of Categories. Categories do not exist independently of the actors that use them. By the same token, category systems, which are the assemblage of categories coordinated in a setting, are a result of multiple actors participating in their institutionalization. Organizations and other market actors, such as intermediaries and raters, are the primary producers and suppliers of categories. These actors influence whether new categories are integrated into, supplant, or exit the category system. As a result, category formation precedes and conditions the phenomena of study in organization and management research, as categories imply that actors identify, assess, and transact with others.

A category is “a class about whose meaning an audience segment has reached a high level of intentional semantic consensus” (Hannan et al., 2007: 69). When “enthusiasts” are convinced and convince others that a category can be defined by certain traits and their members do not share multiple identities, the category becomes accepted and used as a shortcut for identifying and evaluating audiences. As Negro et al., (2010: 14) notes, “the formation of consensus among audiences on which label to apply to sets of producers or offers is the seed of categorization systems.” Category formation therefore lies in an audience’s capacity to agree that a new category is contrasting and is distinguishable from existing categories. Once established, actors use the category as a referent for membership and evaluation (Fleisher, 2009).

Navis & Glynn (2010) in the context of the satellite radio industry developed a different elaboration of how a new category emerges by sanctification by multiple audiences. For them, producers use rhetorical strategies to make both their offering identifiable and distinct from rivals and to make the category appear coherent. They announce and materialize future business implications of the category by using linguistic frames to aggregate a collective identity. For instance, analysts attend to events that manifest the emergence of the category, investors concur in analysts’ decisions, and the media feature the apparition of the new category. Once the category appears institutionalized, producers focus on their distinctive traits, and modulate gradients of membership vis-à-vis the new category to distinguish themselves from rival category members. Analysts become sensitive to inter-firm differences, investors’ behavior becomes specific, and media mentions refer to individualized firms and leaders.

Ruef & Patterson (2009) studied the dynamic relationships between producers and evaluators in examining the basis of industrial category formation. They found that firms needed to belong to crisp categories to obtain bank credit when the industrial classification system became institutionalized, but not beforehand. In the early post-civil war period, the classification system was simple, the local correspondents of Dunn &Company were untrained attorneys, and contestation of the evaluations happened recurrently, therefore firms did not need to belong to crisp categories. During the later period of the late 1880s to 1900, the industrial taxonomy became multi-level and more precise, the evaluators were professional credit reporters, there were few contestations of evaluations, and the category system become effective in disciplining and sanctioning firms that spanned categories. This article is of particular interest because it details the mechanisms that motivate evaluators to discern businesses' creditworthiness as a function of their professionalization and the institutionalization of the nomenclature. It also indicates that category systems are produced and maintained by the actors themselves.

In their review, Durand and Khaire (2017) observe that prior research does a poor job of integrating the various factors that influence category formation. As one of many views, the ecological perspective of category formation does not generalize well across all research contexts because categories are created not only by audience's enthusiasts, but also by producers, media, and critics (Hannan et al, 2007; Negro et al, 2011). On the other hand, the work of producers that ally, and then differentiate their strategies after the category has been accepted by market actors as proposed by Navis and Glynn (2010) or Kennedy (2008), is one other possibility. Finally, the ex nihilo creation of category systems and nomenclatures (as in Ruef and Patterson, 2009 or Espeland and Sauder, 2007) is a relatively rare instance of unbalanced power in favor of raters and intermediaries over candidate organizations. In an effort to reconcile the multiple sources of category formation (from audiences, producers, and intermediaries), Durand and Khaire distinguish two types of category formation. First, category emergence proceeds from "elements extraneous to an existing market. Categories emerge when the existing classification system and categorical structure of markets do not sufficiently account for material novelties sponsored by innovators." Second, category creation occurs where actors delineate new "cognitive boundaries around a subset of elements within a preexisting category system." In their summative effort, the authors describe the nature of novelty, origin, organizational agency, mechanisms for distinction, basis of discourse, legitimacy, and outcomes for each type of category formation. They identify the respective roles of the different actors (audience, organizations, intermediaries) that explain why categories are formed and do or do not gain traction in the system.

Consequences and antecedents of membership in one versus multiple categories.

Independent of the various explanations for category formation, the effects of category membership on organizations follow two lines of reasoning. Categories are lenses by which audiences identify and then evaluate producers and their offerings (Zuckerman, 1999): audiences organize their expectations and evaluations of how well candidates perform along the dimensions that define a category. However, there is disagreement in the research literature on how the processes of identification and evaluation take place. From an ecological perspective, in the case of a new producer or product, audiences automatically recognize the characteristics and boundaries of the new category, positively sanctioning it if it coincides with the prototype and negatively sanctioning it if it does not. For instance, Hsu, Hannan and Kocak (2009) directly associate a producer's grade of membership in a category with an audience's expectations about this category: the lower the grade of membership, the less attention an audience will pay to the producer. Furthermore, the attributions of an audience relative to a producer with multiple category memberships, as in the case of fuzzy identity, coincides with a lower evaluation of competence of the producer (i.e., competence is spread thinner across categories). Therefore, category spanning is negatively sanctioned on the basis of three accounts: 1) the cognitive ambiguity generated by multiple memberships (i.e., it is costlier to process meaningful offerings that belong to diverse classes); 2) the identity ambiguity (i.e., it is more difficult to determine who are the producers that span categories); and 3) the competence ambiguity (i.e., a generalist is supposed to be less proficient than a specialist at the task level).

Using the U.S. feature film industry and the context of Ebay auctions, Hsu et al. (2009) make a case that increasing niche width for a film (i.e., spanning genres) decreases its appeal and box-office success. Moreover, for a potential seller, to identify with multiple labels in describing an Ebay item decreases the odds of an auction's success. Within their contexts, "generalists are penalized not because generalization indicates poor skills, but because generalists are less than full members of the category" (Hsu et al., 2009:166). In an effort to address competence ambiguity, Leung and Sharkey (2013) explicitly adopt measures to control for underlying differences in capabilities across producers that would render category spanners' offerings less appealing or of lower quality. Given a change in a crowd-funding website's policy, they created a matched sample of loans that were listed before and after the removal of category labels. This research design enabled a test of whether a category-spanning penalty would stem from perceptual differences rather than from quality-based differences (which became unobservable after the category labels were removed). They found that before the information was removed, prospective lenders judged a loan as less attractive when its promoters belonged to multiple groups and that the category labels associated with the loan significantly increased the penalty of not finding lenders. After removing the loan labels, the negative effect of multiple membership remained significant, which they interpreted as "strong evidence of the perceptual underpinnings of the multiple-category discount" (Leung and Sharkey, 2013:181).

These studies illustrate that the category-spanning penalty is automatic because it relies on an extremely powerful and common cognitive mechanism: preference for familiarity with existing prototypes (Mervis and Rosch, 1981). While preference for familiarity is omnipresent in everyday life, when dealing with organizations and economic matters, authors argue this cognitive mechanism needs to be complemented by other mechanisms that lead to different forms of categorization processes (Durand and Paolella, 2013; Durand and Boulongne, 2017). These scholars call for greater contextualization of who is categorizing and why, and suggest examining the contingencies of the categorical imperative by introducing alternative approaches, including causal models and goal-based categorization (Schneiberg and Berk, 2010; Durand and Paolella, 2013; Durand et al., 2017).

Responding to this call, the second approach to the processes of identification and evaluation stems from different cognitive mechanisms than automatic association with prototypes, focusing instead on how the attention to and evaluation process can be costly and effortful (Barsalou, 1985; 1991). Two versions of this research coexist. A long existing approach conceives of categorization as an ongoing socio-cognitive process. Porac et al. (1989) contended that organizations often ignore information outside of their category and self-select membership based on their social ties. In this view, categorization is a set of confirmatory information loops that influence decisions and reinforce the existing category system, which can lead to isolation and lock-in of suboptimal strategic paths as the case of the waning Scottish knit-wear industry illustrates (Porac et al, 1989). Glynn and Navis (2013: 1127) underline that categorization is "not purely cognitive but socio-cultural as well because it is anchored in the context in which categorization occurs." There often is an explicit effort on the part of audiences to make themselves understood by producers. For instance, Weber et al. (2008) documented how consumers and some farmers co-elaborated the features of grass-fed meat and dairy products around authenticity, sustainability, and naturalness of cattle products. The convergence of category content ensued through mobilization of consumers to promote the category in multiple social settings and the engagement of producers, resulting in greater membership purity. In another study, because of differences in professional rules, understandings, and definitions, Granqvist and Laurila (2011) described how two audiences (scientists and futurists) have been in open conflict to determine and classify objects in the nanotechnology's category system. These studies have provided a richer birds-eye-view of categorization as an effortful social process, suggesting that multiple category memberships represent transient stages and identity in the making, which may not be automatically penalized. In sum, categories are tools and resources for social actors to interact and elaborate their collective identity in correspondence and in opposition with other social actors.

The other vein of research that views categorization as an effortful social process proceeds from a less constructivist perspective. Broadly speaking, this research suggests that audience members are heterogeneous and make use of distinct theories of value, defined as "how audiences identify issues and solutions, ascribe value, and rank solution providers" (Paolella and Durand, 2016: 333). Referring to market analysts, Zuckerman (1999:1431) seeded the idea that "industry-based category structure is contingent on the prevalence of a theory of value" which pushes organizations to conform to categorical imperatives. To elaborate, this branch of the category literature does not pose that the core cognitive mechanism for identification and evaluation of an entity (an organization, a product) is central tendency, as in the ecological view of categorization. Indeed, central tendency is associated with prototypes and automatic processing of informational cues, and therefore leads to penalizing identity and competence ambiguity. By contrast, the core cognitive mechanism associated with theories of value is conceptual combination, which corresponds to an audience's ability to assemble categorical features in an effort to better achieve their specific needs or goals. When evaluators observe and combine an entity's features to match the idealized functionalities that they seek (Barsalou, 1985; 1991), they value this entity more favorably almost independently of its typicality.

Therefore, for this branch of research on categories, there is no need to require ex ante that prototypes exist and that audiences expect the precise exhibition of prototypical features. In market economies, audience members who search to fit their needs seek out entities that are original, specific, or unprecedented, and thus do not necessarily attend to a typical supplier, but instead to an atypical one. In a longitudinal study of corporate law firms located in three markets (New York, London, and Paris), contrary to the ecological view (Hannan et al, 2007; Hsu et al, 2009; Negro et al, 2011), Paolella and Durand (2016) found that there is a positive association between category spanning and performance of these firms. They explain that for sophisticated, singular, and high-stakes cases, clients find that the more attractive suppliers combine multiple practices. However, this positive relationship becomes statistically insignificant when social evaluation (i.e., rankings of corporate law firms) mediates the direct effect between category spanning and performance. Hence, social evaluation, taken as a proxy for audiences' theory of value, is not a neutral assessment of category spanning. It mediates the relationship between category spanning and performance, and depending on contexts can represent most of the effect between category spanning and performance. In the same spirit, Rhee and Fiss (2014) provide evidence that audiences (stock market in their case) react differently about the same events (announcement of poison pills) depending on the framing around the event and the main speaker making the announcement.

Zuckerman (2017) expands on the idea of categorization as a tool for social evaluation. He contextualizes his seminal 1999 concept of "categorical imperative" to other settings beyond stock markets, a specialized case that is likely an exception rather than the rule. He suggests that in other settings where evaluators have more leeway, play a lesser role, and organizations are more easily distinguished from their products, the discount for category spanning may not be as operative. For example, by comparison stock markets are contexts where evaluators have low freedom, are extremely influential, and organizations are closely tied to their products. Categorization is therefore a tool to realize actors' theories of value, and to make transparent the goals they pursue. A fundamental implication of this revision of the categories perspective is that categorical purity is only one instance among many possible options. In particular, the concept of purity becomes less important if audiences can screen candidates' past performance or assess their status. In sum, the validity of the categorical imperative argument depends on the availability of alternative screens, the usefulness of the category system for the evaluator, and the willingness and competence of candidates to be screened out, among other conditions. Hence, categorization is a useful tool for audiences to identify and value what they prefer to consume, and various conditions can enhance pressures for purity or innovation and unconventionality.

Conformity to and Deviance from categories as strategic behavior. The effect of relaxing previous assumptions that category systems are a given and that actors automatically process informational cues to favor prototypical entities opens "the golden cage of categorical membership norms within which organizations are held" (Durand and Kremp, 2016: 86). This new view has spawned research on categories and strategic decision making that advances well beyond the key concepts of specialism and generalism in ecology theory. In this newer vein of research, categorization becomes an intentional process at the level of actors, for example a producer, an audience member, an intermediary, that pursue personal and situated goals and objectives. Economic valuation and exchanges are therefore more accurately represented in the full light of how actors in markets recombine cues, rearrange meaning, and generate new offerings that upend the existing market order.

While market intermediaries can facilitate transactions by alleviating category ambiguity (Glynn and Lounsbury, 2005; Rao et al, 2005), it is not necessarily beneficial to be a full-fledged member of a category with tight and precise boundaries. Pontikes and Barnett (2015) for example account for the strategic benefit for producers of lenient categories, those categories with blurry boundaries and indefinite meaning. Lenient categories because they are under-specified offer flexibility to producers to modify their strategies as their identity and business model develops. Furthermore, Pontikes and Kim (2017) showed how technology software producers operating under the Gartner ranking system categorize themselves relative to how analysts categorized the same firms. They discovered that both producers and analysts are strategic in their categorizing, but the producers' behavior dominated the interaction while analysts' assessments only weighed in secondarily.

Category age also affects both the likelihood of category spanning and the certainty of whether or not there are penalties for spanning categories. If categories are mature, for example more than 10 years old, penalties are more likely. Hence Pontikes and Kim (2017) demonstrate that category consensus can be quite variable because of the idiosyncrasies of the interplay between the strategies of intermediaries and producers. Earlier, Fleischer (2009) in focusing on intermediaries' evaluation provided rich evidence of why and how conflict of interests generated category ambiguity. She examined the equity rating systems of U.S. brokerage firms prevalent at the end of the 1990s and found that when rating system firms receive fees from candidate brokerage firms, raters do not provide objective information on the market. More precisely, they strategically develop categories with overlapping content and which therefore lack the capability for unequivocal interpretation.

Other research also has taken a social process and social constructionist approach. Kennedy (2008) for example stressed the importance of referring to existing legitimate rivals to carve out a category and to be acknowledged as a new category member. This work and others (Jones et al, 2012; Wry et al, 2011) are important for two reasons: because of its focus on the agentic nature of organizations within the dynamic nature of categorical systems to discipline and sanction; and for demonstrating how actors' normative and cognitive processes can collectively reshape category boundaries, membership conditions, and key outcomes (Kennedy et al, 2010). Along these lines, Hsu and Grodal (2015) present a convincing account of why taken-for-granted product categories open space for producers to decouple their actions from what customers expect. Their study of the light cigarette market provides evidence that nonconformity with the politically correct expectations of reduced tar and nicotine content did not expose cigarette producers to penalties. As the category of light cigarette became institutionalized, it became a cognitive shortcut that consumers recognized and for which they did not make the effort to reevaluate its meaning and characteristics. Eventually, this lack of evaluation provided a lever for tobacco companies to increase the degree of tar and nicotine to comparable levels as those of classic cigarettes.

The most recent research associates category formation with strategy making under various mechanisms. For example, the construction and imposition of labels such as nanotechnology (Granqvist et al 2013), the tactics of category detachment, emulation, and sublimation to make products advance on the status ladder, e.g., the case of Italian grappa (Delmestri and Greenwood, 2016), and the expression of a symbolic narrative around material artifacts (Khaire and Wadhwani, 2010). With the development of research on the strategic possibilities of actors' acknowledgment and detachment from categories and category systems, there is an increase in focus on the idea of "optimal distinctiveness" (see Zhao et al, 2017; Zuckerman, 2016 for a review). In parallel, different types of empirical indicators are being developed that advance the measurement of categorical membership beyond the simpler calculation of distances from average traits (referred to as grade of membership or alignment). Kim and Jensen (2011) refined indicators that hierarchize categorical traits on a *market identity order*, assessing the distance of interspersing conventional (classics) and unconventional opera repertoires. Likewise, Durand and Kremp (2016) offer a measure of *conventionality* that expresses the over-representation of the most common features observed in a given category in the context of U.S. symphonic orchestras' repertoires.

Overall, categorical membership results from multiple interrelated strategic choices: such as positioning product and service offerings in a category system, undermining the existing category system, advancing membership grade in the system or alternatively appearing as (un)conventional, favoring lenient versus crisp categorical membership, leading or following intermediaries' categorization, and so on. All of these strategic choices are compatible with the institutional logics perspective, as we subsequently examine.

II. OPPORTUNITIES FOR FUTURE RESEARCH

Both streams of research share common interests in analyzing the disciplinary mechanisms of markets from a cognitive and normative perspective. The genealogy of the two perspectives reveals similarities in conceptual development and legacies based in their original theoretical formulations. For example, institutional logics work developed in reaction to the limits of studies of structural isomorphism in organizational fields (DiMaggio and Powell, 1983) and two stage models of practice diffusion (Davis and Greve, 1997) to focus on contextual studies of institutional heterogeneity of practice diffusion and cultural meaning systems (Thornton, Ocasio, and Lounsbury, 2012). The categories research began as a reformulation of organizational ecology and thus inherited selective and structural properties of ecological theory, i.e., not being a pure member of a category leads to extinction (Hannan et al, 2007). Logics and categories share common definitions and characteristics, for example logics as patterns of symbolic and material elements used and manipulated by individuals and organizations; and categories as interfaces of cognitive agreement about symbolic and material entities. We take stock of their commonalities, examine compatibilities and discrepancies between the two approaches, and underline how to exploit them in future research.

II.1. Compatible assumptions

The two approaches share compatible assumptions with respect to actors and actors' social and economic environment; we elaborate on these common assumptions by reference to specific studies.

Actor level. At the core of the institutional logics perspective are the concepts of *bounded rationality* and *the structure and focus of attention*, foundational principles of the Carnegie School of decision making theory (Ocasio, 1997; Thornton, Ocasio, and Lounsbury, 2012, chapter 4). Likewise, in category research, rationality is bounded, and stimuli trigger the activation of cognitive mechanisms that make individuals acknowledge and assess entities (Hannan et al, 2007; Hsu et al, 2009). Moreover, institutional logics and categories both embody *semantic content* that actors interpret, an important factor to understanding logics and categories' maintenance and change (Hiatt, Sine and Tolbert, 2009; Lowenstein, Ocasio, and Jones, 2012; Ocasio, Loewenstein, Nigam, 2015). For instance, Dunn and Jones (2010) studied the conflicting vocabulary emblematic of the differences in the care and science institutional logics shaping health care education. Jones and her colleagues pioneered work to measure differences in rhetorical strategies of architects (Jones and Livne-Tarandach, 2008) representing the emergence of a new category, modern architecture (Jones, Maoret, Massa, and Svejonova, 2012). Rao et al (2005) is an example of how categories embody and reflect specific meanings in use; they studied the degree to which the semantic content of words in recipes, not only in terms of the ingredients, but also the culinary techniques, was rooted in one of two categories, traditional versus nouvelle cuisine. They found after fifteen years, the categorical boundary between the two category types became porous, that is the boundary could no longer be used as a yardstick to separate the old from the new.

In addition to sharing assumptions about bounded rationality, the structure of attention, and semantic content, both perspectives also assume that *actors and decision makers are situated*, with consequences for judgment and ability to maneuver (Townley, 2002). For instance, Almandoz (2012) documented that during down market cycles bank founders with a background in the community logic were more likely to follow through and successfully found banks than founders with a finance background, i.e., knowledge and socialization to a community logic versus a market logic produced different founding rates. Pahnke, Katilla, and Eisenhardt (2015) showed how different funding sources associated with different institutional logics, for example corporate venture capital (corporate logic), state funding (state logic), and institutional venture capital (professions logic), had different effects on the innovation of young firms. Because institutional venture capital is organized according to a professional logic, it was more effective in supporting innovation than alternative sources of funding associated with the state and the corporate logic. The argument is partnership networks compared to corporate hierarchies and state bureaucracies can respond faster and are more adaptable to the need for start-up firms to pivot from their original business plans. Likewise, from category research, Durand, Granqvist and Tyllstrom (2017) explicate the dimensions and situated context of the sense-making process of categorization. Actors have needs and interests that influence their use of prototypes and goal-derived categories. The authors stress that the entity to be categorized need not be fixed once and for all, and that the entity exists in both material and symbolic forms that are sensitive to varying interpretations. Thus, the act of categorization varies as well as the context of categorization, for example in public versus private and habitual versus exceptional contexts, and this variation can have significant impact on the antecedents and consequences of categorization processes (Grodal and Kahl, 2017).

A consequence of these similarities is that the logics and the categories perspectives converge on the idea that market actors are not doomed to selection processes, but can actively cooperate and co-work their institutional and categorical environments. For instance, in situations where actors identities are incommensurate, past research on institutional logics demonstrates strategies such as hybridization (Batillana and Dorado, 2010; Pache and Santos, 2013), state mandated truces (Reay and Hinnings, 2009) and logic switching (McPherson and Sauder, 2013) enable cooperation for a better good. These studies suggest remedies for resolving conflicting institutional logics such as giving up one's identification with their home logic or establishing interaction rituals to defer to prominent actors (Jourdan et al, 2017).

Category research also investigates how actors co-build categories under competitive conditions. Wang, Wezel, and Forgues (2016) for example studied the conflict between the hotel's category and hoteliers' reaction to consumers' evaluation to understand the competitive and cooperative pathways that exist between a producer (the hotelier), a platform (TripAdvisor), and consumers. Navis and Glynn (2010) showed how producers cooperate rather than compete in the presence of resource scarcity in order to foster the emergence of the essential traits of a category. Following their insight, Sonenshein, Nault, and Odobaru (2017) in a qualitative study examined how food truck operators created a strategic group identity that involved both cooperative and competitive behaviors. Cooperation was needed to help members achieve the central tendencies of food truck group identity, but competition revealed who achieved the ideal or highest regarded tendencies of the group. This evidence redirects existing theory on competitive dynamics and strategic groups in that incumbent firms helped new members to enter the market, resource scarcity led to cooperation, and competition focused on status within the group, not price.

Historical and social environment contexts. Beyond sharing common underlying assumptions at the actor's level, both perspectives consider historical and social influences primarily as moderators of the main relationships. Institutional theory suggests that through the historical process of institutionalization, the more mature a field, the more the dimensions of institutional logics are likely to cohere and be impactful on individual and organizational behavior (Ruef and Paterson, 2009; Tolbert and Zucker, 1983). This is why longitudinal research designs are preferred in institutional analysis and needed to validate and advance the scope conditions of the two perspectives. Shipilov, Greve, and Rowley (2010) employing a two-stage model provided evidence of the long-term impact of prevailing institutional logics in explaining the adoption of Canadian corporate board practices in a mature field. They found in the second-wave of diffusion, practices were more likely to be adopted by those organizations in which the corporation's logic had prompted adoption in the first wave, the second wave serving to complete a fuller repertoire of the categorical elements of the first wave institutional logic. Note, these findings are contrary to prior neo-institutional research explaining practice diffusion as either for efficiency in the first wave and legitimacy in the second (Tolbert and Zucker, 1983), or mimetic isomorphism and board interlocks in mature fields (Di Maggio and Powell, 1983). As expressed by Meyer and Hollerer (2016: 381) elsewhere: "commitment to a logic-defining concept primes the later commitment to a logic-extending concept."

Historical contextualization reinforces the explanatory power of category systems as well. Indeed, as noted, agents in a mature categorical system are more likely to rely on prototypes and established nomenclatures than on goal-derived categories to determine the value of items and transactions. For instance, Khaire and Wadhwani (2010) describe the genesis of the Indian modern art category. To elaborate the discourse about Indian modern art, agents connected the dots between the characteristics of the local Indian artwork and established criteria of value that were prevalent in the global art world. This connection related the aspirations of potential patrons, such as social climbing and belonging to an enlightened elite, to the traditional features of Indian craftsmanship and cultural symbols. Artists consolidated these connections by integrating the codes of producers in the global art market with selected intermediaries and channel partners to further procure and legitimate identity in these circuits.

As illustrated by these studies, both perspectives account not only for the effects of how historical conditions moderate the effects of logics and categorical systems, they emphasize how social position and social attributes play a major role in how institutional logics are instantiated and category systems evolve. As such, they shed light on how actors retain features and characteristics that constitute the institutional orders and logics that weigh on actors in the future (Clemente et al, 2017). For instance, Quattrone (2015) analyzed over several centuries how the Jesuit order produced large-scale routines and governing principles that coincided with a superordinate unfolding logic. In these selection and retention processes, not all actors are equally endowed with influence and power.

In the category research stream, Sharkey (2014) defends the point that organizations may derive status not simply from demonstrations of quality but also by association with highly-valued membership in certain industry categories. Examining U.S. corporate earnings restatements over a decade, her research demonstrates that investors reacted less negatively to earnings restatements announced by firms which were associated with higher-status industries. Therefore, the consequences of violations differ as a function of the expectations raised by an actor's membership in categories, positional centrality in the social structure, and high or low standards of quality. Likewise, Bowers and Prato (2017) find evidence that changes in category systems in a market lead to displacement in actors' status. New categories introduced by a third party (analysts, raters) generate new and exogenous opportunities to stand out and be socially promoted for an unchanged level of quality, offering some actors unearned status and associated benefits. Finally, among many studies providing evidence of the role of status in how actors instantiate institutional logics, Durand and Jourdan (2012) in studying film producers and financiers established that when the holders of a minority logic were not socialized with the dominant incumbents, their countervailing influences against the incumbents were stronger. As minority logic holders, that is the bankers financing movie projects, increased association with incumbent producers and centrality in the network of film producers, their countervailing influence faded away.

II.2. Complementarities: Opportunities for research

Identifying compatible assumptions between the institutional logics' and categories' perspectives suggests that by ignoring either approach, scholars risk omitting important factors that could contribute and perhaps modify their analyses. This section identifies areas of complementarity and avenues for future research at three levels: the inter and intra variance at the audience level, the formation and recombination processes at the category and logic level, and the actor's identity at the organizational level.

Inter- *and* intra- audience variance. The categories and institutional logics literatures have emphasized how heterogeneous audiences use different categorical systems as cues to form judgments (Pontikes, 2012; Lounsbury, 2007). They often typify actors' expectations relative to a prototype or a dominant logic. On the grounds of inter-audience variance and intra-audience homogeneity, Pontikes (2012) demonstrates that venture capitalists positively view categorical ambiguity whereas clients sanction it. Lounsbury (2007) associates mutual fund types based on differences in an institutional logic, trustee or professional logic versus. Performance or market logic, to the type of contract fund administrators established with independent professional management firms. He finds that the heterogeneity of funds as measured by past strategy and performance, (i.e., growth or non-growth) triggers differences in contracting as a function of the competition between the trustee and performance logics.

Categories research has investigated intra-audience variance. Kovacs, Carroll, and Lehman (2013) for instance studied consumer preferences using more than a million reviews on 18,000 restaurants and found that for the same level of quality, consumers assign higher authenticity ratings to independent- and family-owned restaurants. However, their findings raise the question of how they defined the institutional environment. Is independent and family too narrow a definition for explaining authenticity as a category? The institutional logics perspective suggests that focusing solely on one or two categorical dimensions, i.e., independent and family can bias results because the basis of identity, legitimacy, norms, attention and other elements may be based on a broader set of institutions. If these researchers integrated the institutional logics and categories perspectives in their research design and analysis, it may include, not just professional and family logics, but also "authentic" restaurants that draw on the community logic or the religion logic, such as restaurants, bakeries, and wineries run by various community associations and religious monasteries. Greens in San Francisco run by the Tassajara monks is a well-known restaurant that would be left out of the authors' analysis and as a result may bias findings and reduce the scope conditions of the theory. Another example of the enrichment between the institutional and category research is prompted by Weber et al. (2008). The authenticity, sustainability, and naturalness of the category of grass-fed meat and dairy products is unlikely to occur solely based on the co-elaboration between farmers and consumers. Similar to the example of categorical types of cuisine (Rao et al, 2003), one can argue that issues of the sacredness and purity of food are not just the domain of the beef industry, but also manifest from other or larger institutional environments, in the case of grass-fed meat and dairy products, in community, professional, and religious institutional logics.

Different audience members mobilize different categories and members' ability to identify categories stems from possessing different theories of value to motivate and explain needs and aspirations (Zuckerman, 1999; 2017; Paolella and Durand, 2016). Scholars acknowledging this point also add that audiences can be constituted by multiple theories of value and that members from different audiences can share common theories of value (Durand and Boulongne, 2017). For example, some pro-environment consumers and investors may share the same theory of value while not being part of the same audience type. In theory, they may observe identical institutional logics despite being typed as belonging to different audiences: for instance, both investors and clients may value a community logic over a corporate logic. As research on organic food and nanotech categories illustrates (Granqvist and Ritvala, 2016), producers as well as public authorities and audiences may have convergent interests in having a goal-based category imposed to set the stage for sane competition. If no clear prototype emerges to stabilize market expectations, producers and audience members may disengage. While the category research to date has not ventured into the particularization of individual theories of value within audiences, evidence suggests that audience homogeneity exists across audience types (Hahl, Zuckerman, and Kim, 2017). Sharing and instantiating the same institutional logic is a potential explanation for this inter-audience homogeneity. Hence, a first complementarity between the two research streams at the level of audiences indicates that the institutional logics framework may explain differences in theories of value within an audience and similarities in theories of value across audiences.

As noted previously, early studies in the institutional logics perspective focused on the influences of a dominant logic (Thornton and Ocasio, 1999). Hence, by stressing the prevailing logic, there has been potential to erroneously pre-fixing audience properties and therefore potentially misunderstanding the existing degrees of heterogeneity and agency in the population. For instance, in their study of organizational resistance, Marquis and Lounsbury (2007) assume that out-of-town and generalist banks are all likely to instantiate a performance logic as distinct from local and community banks. Yet, the institutional logics perspective does not assume that individuals have or use the same valuation criteria across institutional orders, nor have the equal capacity to recognize institutional contradictions (Battilana and Dorado, 2010; Voronov and

Yorks, 2015). The intrinsic coherence of an institutional logic is not an iron cage. The categories perspective provides the mechanisms explaining why some features and elements are more noticed, salient, and valuable for audience members (Paolella and Durand, 2016; Smith and Chae, 2017) and therefore offers a contributive addition to the institutional logics perspective.

Therefore, we argue that associating the duel concepts of dominant and multiple institutional logics and category systems and references used by audiences can specify a richer set of the assumptions and scope conditions for both perspectives. Institutional logics provide a set of frames that specify the criteria that guide how and why audience members express theories of value. Hence, there is great potential to advance categories scholarship by using the institutional logics comparative subsystem framework to classify theories of value for multiple audiences. Reciprocally, applying categories theory to institutional logics research provides a new way to explain why some actors may be more capable than others in handling multiple conflicting institutional logics and engaging in logic switching and recombination. The question may be a function of either how much an actor values one categorical element over another or which cognitive mechanism the actor uses (i.e., central tendency, conceptual combination) in a given situation. Future research geared toward understanding assessment, likeability, preferences, adoption and any action resulting from valuation can likely benefit from acknowledging the coexisting influences of category systems and institutional logics on the antecedents of the outcomes under study. The crux of this problem is to account for both inter-group homogeneity and intra-group heterogeneity in theories of value, without committing inextricable particularization such that findings do not generalize.

Logic and Category Levels: Formation and Recombination. Research suggests that category emergence proceeds from "elements extraneous to an existing market" (Durand and Khaire, 2017). Moreover, changes in institutional logics lead to new category creation and to new meaning of existing categories (Ruef, 1999; Ocasio and Joseph, 2005). Yet, the few studies that connect these two threads of knowledge truncate the analysis to the later stages of institutionalization and miss the opportunity to connect the dots on the origin of the category itself. For instance, Rao, Monin and Durand (2003) in their study of the categories of classic and nouvelle cuisine, argued that social identity movements explained the shift in categories in the culinary profession. The authors identified four mechanisms linked to this change: the sociopolitical legitimacy of food critics as activists, the theorization of new roles, the prior defections by peers and gains to peers, and gains to defectors as identity-discrepant cues. Note the mechanisms the authors identified do not foreshadow why the social identity movement led to the emergence of the nouvelle cuisine category. It is not obvious that such a movement, which upended the theatre, fiction books, and movie fields could also revolutionize the culinary field, a distant field on many accounts (e.g., manual versus intellectual, imitative versus innovative, collective versus individual creation).

By comparison, Jones, Maoret, Massa and Svejenova (2012) investigated the formation of a new category, rooting its account in an analysis of institutional logics. They used network analysis of actors' vocabulary to identify institutional logics and how these logics shaped the creation of a new category called modern architecture. By examining the centrality of key words and their relation to material practices from 1870 to 1975, they delineated two variants of a professions logic, modern organic, associated with the analog of nature, and modern functional, associated with the analog of industry. Jones et al. (2012) provide evidence of how architects embodying these two types of institutional logics battled over the definition and boundaries of the category of modern architecture. The comparison of these two studies suggests that knowledge of the instantiation of institutional logics can inform the analysis of category emergence and creation.

Research on the decomposability of an institutional logic into categorical elements (Thornton, Ocasio, and Lounsbury, 2012; Battilana and Lee, 2014) offers another avenue for better understanding how and why the content of an institutional logic evolves and its use changes over time. Through the processes of transposition and reinterpretation, actors are capable of assembling categorical elements from different institutional orders to modify an institutional logic. For instance, Purdy and Gray (2009) describe the development of dispute resolution practices over two decades, identifying how actors engaged in four processes (transformation, bridging, grafting, and exit). The category research can advance understanding the elaboration and change in institutional logics beyond description of these processes by uncovering the general mechanisms for how and why decision makers orient their organization in one direction or another. For example, understanding how audience members select reference groups to establish their comparisons (Smith and Chae, 2017), how they assess category valence and combination (Vergne, 2012), and whether they find value in enforcing crisp versus lenient categories (Leung and Sharkey, 2013; Pontikes and Barnett, 2015) provide fodder to a description of the process at play in institutional logics' evolution.

In sum, research is needed to uncover the regularities explaining why, how, and when new categories emerge from adjacent or more distant institutional logics; who are the actors that bridge across institutional logics; and the reasons emergent categories pertain to the prevalent category system. The same questions about category creation, initiated by incumbent organizations seeking to regenerate the conditions that make their theory of value preeminent in a market (Durand and Khaire, 2017) can be addressed by the institutional logics' lens.

Accordingly, the categories approach brings new light to the reasons why certain actors assemble and rearrange dimensions of logics. Categories research provides concepts (e.g. prototypes, goal-based categories, contrast, spanning, ambiguity, leniency, grade of membership), mechanisms (e.g., central tendency, familiarity, conceptual combination), measurements, and relationships that complement the explanation of why hybrid organizations strive and decay beyond the more traditional institutional explanations rooted in identity and legitimacy and other social processes and cultural accounts. Therefore, future research targeting the first stages of category formation and the recombination within and across institutional logics will benefit from a greater integration of the two literatures.

Actors' identity: Both perspectives have a theory of organizational identity based in contrast effects, i.e., contrasting focal organizational traits and characteristics relative to predominant categories and alternative institutional logics. Both perspectives argue that decision makers span and stretch categories and logics selectively and strategically (Battilana and Dorado, 2010; Kim and Jensen, 2011; Durand et al, 2013; Pontikes and Kim, 2017). While the two perspectives complement each other on what makes an organization's identity coherent and conform to expectations or at odds and deviant, the two perspectives have not been used much in concert to study organizational identity, a reason being that while complementary, their approaches begin at different origin points.

The categories literature adheres to a bottom up process in which product characteristics aggregate up to the organization to determine organizational identity. For instance, Kim and Jensen (2011) aggregate the different opus performed by opera houses to characterize market identity and the corresponding expectations of audiences. Hsu (2006) compounds multiple movies produced to typify whether or not a producer is a generalist or specialist. However, doing so fails to account for how organizations instantiate logics that audiences recognize and comprehend from the larger institutional environment, i.e., independent of the aggregation of product features. This bottom-up method ignores well established theory and empirical findings in neo-institutional theory in which organizations are inscribed by generalized facts prevalent in their larger institutional environments (Meyer and Rowan, 1977; Anteby, 2010; Jourdan et al, 2017). And this inscription may be independent, if not inconsistent with identification processes stemming from the aggregation of observable cues at lower levels of analysis –such as products, repertoires, or business portfolios (Negro et al, 2010; Kim and Jensen, 2011; Smith, 2011; Paolella and Sharkey, 2017).

While in theory not limited to only top down processes (see Thornton, Ocasio, and Lounsbury, 2012 chpt 4), many studies in the institutional logics literature do not account for the bottom-up aspects of audiences. Rather, institutional logics research emphasizes the more generalizable dimensions of logics, focusing on whether organizations respect them or not (e.g., bases of identity, authority, governance, etc.) as illustrated by the case of business actors involved in production of cultural products (Rao et al, 2005; Jones et al, 2012; Shymko and Roulet, 2017).

One potential advance with promise is to capture both bottom up and top down aspects of organizational identity. The categorical elements that form the basis of organizational identity in institutional logics research are not well developed. The aggregative measures used in categories research could be easily transposed into the institutional logics perspective to express with more fine-grained attention how organizations instantiate the multiple categorical elements of a logic. Moreover, the sources of legitimacy, identity, authority, etc., are likely to vary by different classifications of actors. Not all actors even within an institutional order may share a common sense of legitimacy, identity, and authority, nor use the same vocabulary—these categorical elements may vary to differing degrees for example by different audiences and producers as a function of their receptivity to various normative constraints imposed by an institutional logic. Exploiting this complementarity is therefore promising in elaborating this macro to micro and micro to macro interaction to define organizational identity, why it changes over time (in the absolute and for a specific audience), and what consequences this entails for the organization's choices and outcomes.

II.3. Differences: Opportunities for Research

We underlined the compatibility of assumptions across the two research streams and some promising complementarities. However, some striking differences between the two research fields are worth mentioning as they could appear as limiting the prospects of further integration; namely: each theory is more developed at a different level of analysis; the two perspectives account for conflict differently; and use different methods of analysis and approaches to data. These differences provide opportunities for future research.

From Theory Level Differences to Boundary Conditions. An obvious and consequential difference between the categories and institutional logics literatures is the level of analysis at which each theory has developed. Most of the categories research is meso-level theory; i.e., an intermediate level that consists in organization-to-organization or organization-to-individual level theorizing. The institutional logics perspective is a meta-theory that encapsulates institutional orders, logics, organizations, and individual behavior. It is often integrated with other meso- and lower-level theories and concepts to produce meaningful refinement in scope conditions of existing theorizing, for example theories of attention (Thornton & Ocasio, 1999; Thornton 2001, 2002), paradox (Jay, 2013), dynamic constructivism (Glaser et al, 2016), symbolic interaction (Jourdan, Durand, & Thornton, 2017), political coalitions (Greve and Zhang, 2017), ecology (Marquis and Lounsbury, 2007), agency theory (Chung and Luo, 2008), path dependence (Shipilov, Greve, and Rowley, 2010), and cultural sociology (McPherson and Sauder, 2013; Battilana and Dorado, 2010). This asymmetry has implications for further integration to expand the capacity for multi-level theory and empirical research.

One view on this asymmetry is that the explanatory power of each theory depends on conditions of the other. Categories are a basic unit of cognition and a conduit through which institutional logics shape cognition (Thornton and Ocasio, 2008). Thus, the categories perspective is conceptually connected with institutional logics' theory and can contribute to enrich it. As previously expressed, research on categories provides theoretical concepts and mechanisms (Jones et al., 2012; Durand and Boulongne, 2017; Jancsary, Meyer and Hollerer, 2017) that explain how institutional logics are identified as available to individuals and organizations to access and activate. The concepts, theoretical mechanisms, and stylized facts brought about by the categories research (consistently replicated results as reviewed in section I.2) help to better identify the basic organizing principles and expectations for the institutional logics' perspective: after several decades of research on categories, we know the reasons why central tendency or conceptual combination prevails, when actors employ analogies or refer to ideals instead of prototypes, how category system's properties (contrast, inclusiveness) and actors' characteristics (status, expertise) influence the main relationships of interest. This knowledge dovetails well with the institutional logics perspective and could further develop its scope conditions and explanatory power.

Reciprocally, how actors constitute category systems, mobilize categories, and evaluate entities depend on which institutional logics are salient, invoking some interpretations of the situation, but not others (Lounsbury, 2007; Quattrone, 2015). Notably, two views in the categories literature, the goal-based approach (Durand and Paolella, 2013; Granqvist and Ritvala, 2016; Zuckerman, 2017) and the socio-constructivist view (Granqvist et al, 2013; Grodal & Kahl, 2017) suggest advances by using the institutional logics perspective to anchor cognition in an institutional context. Therefore, to explain why actors mobilize certain categories, which categorical features are more meaningful, what degree of ambiguity actors can tolerate, scholars need to understand and account for the presence and salience of some institutional logics over others. Hence, there are situations where category theory precedes over institutional logics theory in explaining phenomena, notably when there is no conflict in norms and values in a situation. Likewise, when prototypes of the category system are established and actors are identified, category theory may have low explanatory power, while the institutional logics perspective probably accounts for most of the variance of the phenomena under study. In the middle case, both theoretical perspectives likely account for a significant share of the explanation. Overall, the differences in level of theorization which are revealed by an integrated approach represent an opportunity for a clearer delineation of the validity of both research perspectives.

Role of conflict. Scholars have highlighted the role that institutions play in legitimating and sustaining "common cognitive understandings" e.g., as expressed as labels, vocabularies, and categories (Cornelissen et al., 2015: 14). As previously reviewed, characteristics of the institutional context trigger activation of cognitive mechanisms by reference to linguistic and visual properties that connote norms and values associated with the situation (Glaser et al. 2016). However, considerable research on categories does not account for these nuanced connotations and how they may conflict with other interpretations (particularly the branch of category studies rooted in prototype research). Most of the prototype research is limited to examining informational signals that trigger cognitive mechanisms (Hsu, 2006; Hsu et al, 2009; Negro et al, 2011). For instance, Wry, Jennings, and Lounsbury (2014) focused on the effects of the 'header-modifier,' i.e., how a key word or lead term as in 'cat house' or 'mobile phone' modifies cognitive perceptions. They found that depending on how the 'header-modifier' applies, some new ventures will be more likely to be funded than others. Notably, when a scientific venture combines commercial acumen, it is more likely to be funded than when a commercial team strives to associate with scientists. This linguistic property triggers cognitive mechanisms and actions; however, it is not embedded in a broader institutional context where values may be more or less subject to conflict.

Recent studies on market categories examine moral and value-laden features more susceptible to igniting discord among audiences (Vergne, 2012). These studies broaden the definition of a goal to include the underlying "ideals" of the goal, for example by focusing on environmental footprints and human rights protections as features of a category. For instance, the research on socially responsible investment (SRI) funds shows how differences in goal ideals differentiate the ability of asset managers to compete in shaping a new category (Arjalies, 2010). With the idea of moving this research beyond attentional stimuli and consonant or discrepant cues, what if the categorical imperative and related categorical principles were analyzed from an institutional and social normative perspective? Here, there is interesting research to build on to better understand the value-laden features of market categories and their potential impact on social processes that affect theoretical mechanisms and scope; for example, with the research on practice and proscribed exchanges (e.g., cadaver trade in Anteby, 2010) and the institutionalization of illegitimate products (e.g., erotic movies in the 1970s in Jensen, 2010). In a rare study tackling the idea of normative and political conflict and domination, Quinn and Munir

(2017) identify how a hybrid category (impact investing) led an organization to establish and perpetuate unequal power relations in exploiting natural resources. They studied the case of forest exploitation in Sub-Saharan Africa (Uganda) and land disputes between a portfolio company from a UK-based impact investment fund and local communities. In analyzing the imperatives weighing on the fund's activity reporting, and the subsequent actions it undertook to maximize its impact, they state: "as powerful organizing devices, categories are political and cultural productions fraught with contention" (Quinn and Munir, 2017:115).

A key difference between the categories and institutional logics research is that the logics perspective has dealt with normative and value conflicts (e.g., Friedland and Alford, 1991; Thornton and Ocasio, 1999). Institutional logic scholars have integrated research in cognitive and social psychology and have studied (at least) three approaches that are relevant to this discussion. First, Nigam and Ocasio (2010) revealed how attention and sense making are triggered by events in the institutional environment in the context of sensitive political issues. Second, Suddaby and Greenwood (2005) showed how actors exposed contradictions in institutional logics to gen-up a rhetorical narrative for and against a new organizational form. And third, how some actors, but not others are able to express themselves to make some logic categorical elements more salient, attended to, and convictional than others: McPherson and Sauder (2013) evidenced how court room professionals "hijack" institutional logics of their opponent to make the court room docket progress efficiently. They found that of all the professionals, the psychologists were significantly less likely to switch and take the logic of the other in negotiating the case compared to the lawyers and probation officers. This suggests that not all logics are as available and accessible to any actor and that logics are not neutral in how cognition is activated; some logics are more malleable to moving and switching in different contexts, while others are intractable.

The idea of examining both the cognitive mechanisms and the institutional context concurrently, not in isolation, in future research is therefore central to understanding the transposition of categorical elements of institutional logics. For the institutional logic literature, it contributes to understanding the (cognitive) limits of actors to use institutional logics as a cultural tool kit. Theories of value can be conceptualized as orthogonal to the vertical institutional orders and explain the categorical elements of values and behaviors in an industry, market, and other environments. Reciprocally, for the category literature, institutional logics embed actors in a socialized and institutional context, paving the way for broadening the investigation and inclusion of more contentious categorical features in the analyses. For the categories research, the institutional logics perspective identifies different valuation criteria, how categories fit together in a system of meaning, and how these systems conflict and can find ways of resolution; for example via resistance (Marquis and Lounsbury, 2007; McDonnell and King, 2013), deference (Jourdan et al, 2017), rhetoric and other discourse tactics (Suddaby and Greenwood, 2005). The tight and loose coupling of categories within an institutional order can indicate the weakness and stability of categories of an institutional logic and its level of resilience to replacement by another logic. It can show the social and political influence on vocabularies and provide a theory of regulation and change of category definitions (Lowenstein, Ocasio and Jones, 2012), all phenomena that deserve more research.

Methods: Better Measures for Better Findings. Another striking difference between the two perspectives consists in the type of methods employed. The research on institutional logics aims at understanding the effects of multiple institutional logics over longer periods of time (Dunn and Jones, 2010; Jones et al, 2012), whereas the category research draws on cross sectional and relatively shorter time periods, i.e., on the spot evaluation or choice, or a few years of observation, a decade at most. As noted, because research on institutional logics covers longer periods of time, it is able to unearth the origins and varied meanings of categories, the breadth of categorical boundaries, and the true (or not) challenges of hybridization and boundary spanning (Battilana and Dorado, 2010; Quinn and Munir, 2017; Paolella 2017). This longitudinal and more dynamic research design complements the categories literature which remains largely focused on settings where category systems are fixed or where there is a simplification of the institutional context, such as in a study of one industry or a product category.

Researchers in the categories literature have made strides toward conceptual and methodological refinement in understanding the multiple combinations of features that define a category. Authors like Carroll, Hannan, Hsu, Kocak, Negro and Pontikes have no doubt built rich and reproducible techniques to assess categorical purity, contrast, and spanning effects, among other traits. Paolella and Durand (2016) have attended to the vertical structure of category systems and suggested the construct of "inclusiveness" which represents the co-occurrence of n-uplets of categories (duos, trios, etc.) in the entire universe of all observable data. For instance, the association of some duos (practice A and practice D) could be less represented than some trios (A, B and C) in the entire set of n-uplet offerings present in a given year in the market.

The operationalization of these more advanced methods in the categories literature can enrich the institutional logics research. Institutional logics research has used case studies to describe and infer relationships among the constitutive elements of a logic and the way that an actor instantiates the logic. While there are studies that provide evidence of the processes of the recombination of logics' elements that lead to hybridization, comparatively speaking their identification and operationalization are not at the same level of refinement and systematic analysis as in the categories literature. Furthermore, this lack of precision could lead to under-specification of each organization's or audience's particularities under consideration. Hence, we advocate for the development of multi-method research. Alongside cases and qualitative evidence, large longitudinal datasets provide a sound approach to investigate the association between factors of interests (e.g., panel-data regression analysis). Other investigations can complement them, i.e., specific forays using other appropriate techniques in order to probe causation between factors. For instance, identification techniques such as instrumented variables, matched sampling, and difference-in difference strengthen the estimation of directional effects. At the finer levels of analysis, when approaching individuals in their capacities to apprehend categories and institutional logics' elements, randomized controlled trials, experimental designs, but also text mining techniques to name a few are of value to probe the attentional, cognitive, and linguistic mechanisms that actors activate in specific situations. In sum, future research should gain from associating the two approaches not only at theoretical and phenomenological levels, but also by combining their methodological strengths.

Conclusion

This review has motived the benefits of integrating knowledge, concepts, and methods from the categories and institutional logics literatures. These two fields have not been integrated before for reasons that wane nowadays. First and foremost, as noted previously, the intellectual roots of the logics and categories perspectives stem from two theoretical traditions in organization and management studies that textbooks long viewed as distinct, institutional and ecological theories. It has taken time for research to dispel unproductive questions and debates this theoretical bifurcation encouraged, such as whether or not organizations adapt (institutional theory) or are selected out of the population (ecology theory). Second, as expressed above, there exist differences in methods' conventions and sophistication between the two research traditions. As more cross-over scholars ignore these differences and academic demands for evidence increase, multi-method research proves advantageous to circumscribe the factors of influence and their direction. Third, under the steam of new generation scholar cohorts, the dividing lines between the neo-institutional and organizational ecology scholarship legacies atrophy. Moreover, with the continuing professionalization of field of business strategy new opportunities are available to apply and merge knowledge of institutions and categories in market settings. This paper materializes the convergence between these two streams, and the fertilization of current and new fields it begets.

The first part reviewed the institutional logics research, starting with the influence of institutional logics on organizations, the way organizations deal with multiple institutional logics, and how they mobilize them to induce institutional change. The review of the categories research was organized around the formation of categories, the consequences and antecedents of belonging to one versus multiple categories, and conformity and deviance from categories as strategic behavior.

The second part of the review focused on the compatibility of the underlying assumptions of the two literatures with the intent to lay the ground work for identifying the complementarities between the two perspectives that may suggest opportunities for future research. Compatible assumptions were identified and discussed at the actor level, in terms of bounded rationality, the structure and focus of attention, semantic content, and decision makers as situated; and at the historical and social environmental context (mostly) as moderators of main effect relationships. Table 1 synthesizes our presentation.

Insert Table 1 about here

Areas for future scholarship gleaned from these complementarities included, inter and intra audience variance in which the crux of many of the issues highlighted in the review suggest the need for better theories of value to explain inter-group homogeneity and intra-group heterogeneity. One of the clear benefits of integration of the two perspectives gleaned from the reviews is that institutional logics can help explain differences in theories of value within an audience and similarities in theories of value across audiences. Future research along these lines holds promise to correct the over-generalization of the categorical imperative across markedly different institutional contexts, since market analysts' theory of value has strict boundary conditions (Zuckerman, 2017). Research has shown that different audiences mobilize different categories and identification mechanisms which likely proceeds from each audience possessing a different theory of value, i.e., a theory of value motivates and explains the needs and aspirations of audience members (Paolella and Durand, 2016). In theory, the categorical imperative will likely vary depending on which institutional logics are salient in the particular context, a question yet to be examined empirically. While the categories research has been applied in many market-based settings, the review of the institutional logics research highlights that markets are only one institutional order in society and market based practices are often influenced by non-market forces (Greenwood et al. 2010), suggesting an opportunity to expand the scope conditions of category theory.

The categories research can measurably add to the institutional logics perspective by helping to define the illusive concepts of legitimacy and identity, which are central to any institutional logic. Both the concepts of legitimacy and identity have been difficult to measure, perhaps because of their taken-for-granted importation from neo-institutional theory. The categories research argues that legitimacy stems from categories' characteristics and identity from the degree of membership in categories, and provides methods to measure and test their effects. More importantly, there has been slow progress in developing new generalizable concepts and employing more sophisticated methods of analysis in the institutional logics perspective. Category researchers developed concepts and theoretical mechanisms that can be immensely useful to solve this problem: for instance, central tendency in association with prototypes, grade of membership, fuzzy identity, and category spanning penalty; but also, goal-based categorization, conceptual combination, theory of value and conventionality. By comparison, current measures of institutional logics are crude, e.g., qualitative case studies, historical time periods, geographical regions, actors' functional backgrounds, excepting the newer content analytic research on vocabularies (Jones et al., 2012), the priming of logics in experimental designs (Glaser et al, 2016), and the techniques of visual imageries and semantic networks (Meyer et al. 2016).

In sum, three types of consequences emerge from analyzing the findings of this review. Failing to span these literatures will stall both perspectives due to unspecified mechanisms, processes, and contexts. On mechanisms, institutional logics can provide the groundings for actors' theory of value for categories research and categories can help institutional logics researchers define the categorical elements of the inter-institutional system that underlie concepts such as the sources of legitimacy and identity. On processes, the institutional logics perspective can help explain the emergence, creation, and change in the meaning of categories. The categories research can help explain how institutional logics become instantiated in a market. On context, categories researchers are interested in inter and intra audience variance and expanding their ability to define and assess a fuller set of influences of the institutional environment. The institutional logics perspective can help to specify the heterogeneity of the institutional environment, particularly outside the market context. The categories research can help institutional logics researchers identify the distinctions between the producers and audiences of institutional orders and their evaluation proclivities.

Overall, the mutual enrichment and cross-fertilization between two fast-growing streams of research in organization and management studies promises to reveal unobserved and previously unobservable sources of explanation to multiple sociological, political, and economic phenomena.

Table 1. Institutional Logics and Market Categories: Compatibilities, Complementarities, and Differences

Compatible Assumptions	
Actors	Bounded Rationality Structure & Focus of Attention depend on cues and stimuli Actors & Decision Makers are situated Actors influence material and symbolic elements of the inter-institutional and market category systems Competitive Conditions result from actors' positioning vis-à-vis institutional logics and market categories
Contexts	Historically Contingent: importance of context maturity to explain phenomena Socially Contingent: importance of social structure and actor position to explain phenomena
Complementarities	
Audience Variance	IL contributes to CL by explaining the origins of audience members' theories of value. CL contributes to IL by explaining how and why actors pay attention to cues and signals and the resulting interpretations and actions. Integration benefits: more comprehensive explanations of inter- and intra- audience heterogeneity and resulting behaviors
Formation and Recombination	IL contributes to the explanation of how and why categories form and category systems evolve CL provides concepts and mechanisms that contribute to explaining the presence, reinforcement, and waning of IL's constitutive elements Integration benefits: explaining category formation and the recombination within and across institutional logics
Actors' Identity	IL contributes to CL by theory and empirical evidence of top-down processes shaping organization identity (influences of legitimacy, authority, etc.) CL contributes to IL by theory and empirical evidence of bottom-up processes that constitute organizational identity (aggregation of product features) Integration benefits: micro to macro and macro-to-micro processes of actors' identity's constitution, change, and outcomes
Differences	
Levels	IL is a 'meta' theory and CL is a 'meso' theory Benefits of integration: identifying the conditions of the theoretical and empirical validity of each research perspective and reinforcing their explanatory power
Role of Conflict	IL theorizes conflict as an explanatory mechanism, CL has mostly ignored conflict

	Benefits of integration: accounting of the cognitive mechanisms associated with conflict and expansion of research scope to product and market settings influenced by values, ideologies, and normative principles.
Methods	IL uses case studies and longitudinal correlational studies. CL uses multi-settings, correlational, but also experiments, matched samples, and causal identification techniques Benefits of integration: Advocating for multi-method studies where correlational studies are complemented with at least another method (e.g., qualitative, textual, computational linguistics, causal identification, etc., depending on research questions and study design issues).

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